

COURSE SYLLABUS

Advanced SQL Server Performance Tuning

Software Tutorial Services LLC

COURSE NUMBER

65318A

DURATION

2 days · 6 modules

AUDIENCE LEVEL

Advanced

FORMAT

Instructor-Led / Self-Paced

Course syllabus

COURSE OVERVIEW

This 2-day course builds on the SQL Server 2025 query analysis skills taught in 65317A and focuses on real-world troubleshooting techniques. Students will be taught to tune and optimize queries using SSMS 22 and other tools on SQL Server 2025.

AT COURSE COMPLETION

After completing this course, you will be able to:

Analyze query execution plans to identify performance bottlenecks in SQL Server 2025

Apply real-world tuning techniques using SSMS 22 and advanced diagnostic tools

Optimize query performance through index strategy and execution plan optimization

Troubleshoot and resolve performance issues in production SQL Server environments

Leverage SQL Server 2025 new features and Microsoft tools for continuous performance monitoring

PREREQUISITES

- Course 65317A - Querying and Analyzing SQL Server 2025, or equivalent knowledge
 - Solid understanding of T-SQL and query fundamentals
 - Experience working with SQL Server in production environments
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COURSE OUTLINE

MODULE 1 Performance Troubleshooting Methodology

140 min

Establish a systematic approach to identifying and diagnosing SQL Server performance problems. Learn to establish performance baselines and apply a consistent troubleshooting workflow.

LESSONS

Performance Problem Identification
Systematic Troubleshooting Workflow

LAB EXERCISES

Lab 1 – Identifying a Slow Query

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

Identify common SQL Server performance problems using diagnostic queries and tools
Establish performance baselines for proactive monitoring
Apply a systematic troubleshooting workflow to diagnose performance issues
Analyze workload statistics to locate inefficient queries

MODULE 2 Advanced Index Tuning

140 min

Master index optimization strategies including missing index identification, maintenance strategies, and fragmentation management to improve query performance.

LESSONS

Missing Index Analysis
Index Fragmentation and Maintenance

LAB EXERCISES

Lab 2 – Index Optimization

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

Identify missing indexes using SQL Server 2025 recommendations and DMVs
Design and implement optimized index strategies for complex workloads
Evaluate index fragmentation and apply appropriate maintenance strategies
Test and validate query performance improvements after index optimization

MODULE 3 Query Store Performance Diagnostics

140 min

Leverage Query Store capabilities in SQL Server 2025 to analyze historical query performance, identify regressions, and stabilize execution plans.

LESSONS

Query Store Architecture
Plan Forcing and Stability

LAB EXERCISES

Lab 3 – Using Query Store for
Troubleshooting

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

Describe Query Store architecture and configuration options in SQL Server 2025
Analyze historical query performance data to identify trends and regressions
Identify queries with performance regressions using Query Store reports
Force optimized execution plans to stabilize query performance

MODULE 4 Wait Statistics and Resource Bottlenecks

140 min

Diagnose system-level performance problems by interpreting wait statistics and identifying CPU, I/O, and memory bottlenecks in SQL Server 2025.

LESSONS

Wait Statistics Fundamentals
Resource Bottleneck Mitigation

LAB EXERCISES

Lab 4 – Diagnosing Resource
Bottlenecks

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

Interpret wait statistics to identify the root cause of performance problems
Diagnose CPU bottlenecks and implement mitigation strategies
Identify and resolve I/O bottlenecks affecting query performance
Recognize and address memory pressure issues in SQL Server 2025

MODULE 5 Concurrency and Blocking

140 min

Troubleshoot concurrency issues including blocking chains, deadlocks, and isolation level impacts on SQL Server 2025 workloads.

LESSONS

Locking and Blocking Fundamentals
Isolation Levels and Concurrency

LAB EXERCISES

Lab 5 – Blocking and Deadlock Investigation

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

Identify and analyze different lock types and their impact on concurrency
Diagnose and resolve blocking chains affecting transaction throughput
Capture and analyze deadlock graphs using SQL Server 2025 tools
Apply appropriate isolation levels to balance concurrency and consistency

MODULE 6 Advanced Query Optimization

140 min

Master advanced optimization techniques including parameter sniffing troubleshooting, plan stability, memory grants, and adaptive query processing in SQL Server 2025.

LESSONS

Parameter Sniffing Deep Dive
Memory Grants and Plan Stability

LAB EXERCISES

Lab 6 – Parameter Sniffing Troubleshooting

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

Diagnose and resolve parameter sniffing issues affecting query performance
Implement strategies for maintaining plan stability in dynamic workloads
Analyze and optimize memory grant issues causing performance problems
Leverage adaptive query processing features in SQL Server 2025